

# MTH-645 Optimization Techniques

## Assignment (2)

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# 1

Carry out three steps of the conjugate gradient method to find the minimum point of the functions:

$$(i) F(x) = 4x_1^2 + x_1x_2 + 2x_2^2 - 19x_1 - 14x_2$$

$$(ii) F(x) = x_1^2x_2^2 + (x_1 - 3)^2 + (x_2 - 2)^2.$$

Start at  $x = (0, 0)$  and use calculus to find the minimum in each conjugate direction.

Comment on your result after the three steps in comparison with the actual minimum point.

## Solution:

We'll use the Fletcher-Reeves algorithm, which is given by:

### **Algorithm 5.4** (FR).

Given  $x_0$ ;

Evaluate  $f_0 = f(x_0)$ ,  $\nabla f_0 = \nabla f(x_0)$ ;

Set  $p_0 \leftarrow -\nabla f_0$ ,  $k \leftarrow 0$ ;

**while**  $\nabla f_k \neq 0$

    Compute  $\alpha_k$  and set  $x_{k+1} = x_k + \alpha_k p_k$ ;

    Evaluate  $\nabla f_{k+1}$ ;

$$\beta_{k+1}^{\text{FR}} \leftarrow \frac{\nabla f_{k+1}^T \nabla f_{k+1}}{\nabla f_k^T \nabla f_k}; \quad (5.41a)$$

$$p_{k+1} \leftarrow -\nabla f_{k+1} + \beta_{k+1}^{\text{FR}} p_k; \quad (5.41b)$$

$$k \leftarrow k + 1; \quad (5.41c)$$

**end (while)**

i.  $F(x) = 4x_1^2 + x_1x_2 + 2x_2^2 - 19x_1 - 14x_2$

$$F_{x_1} = 8x_1 + x_2 - 19$$

$$F_{x_2} = x_1 + 4x_2 - 14$$

$$\nabla F = (8x_1 + x_2 - 19, x_1 + 4x_2 - 14).$$

iteration 1:

$$x_0 = (0, 0)$$

$$f_0 = 0$$

$$\nabla f_0 = (-19, -14)$$

$$p_0 = -\nabla(f_0) = (19, 14)$$

$$\begin{aligned} F(\alpha_0) &= 4(19\alpha_0)^2 + (19\alpha_0)(14\alpha_0) + 2(14\alpha_0)^2 - 19(19\alpha_0) - 14(14\alpha_0) \\ &= 1444\alpha_0^2 + 266\alpha_0^2 + 392\alpha_0^2 - 361\alpha_0 - 196\alpha_0 \end{aligned}$$

$$= 2102\alpha_0^2 - 557\alpha_0$$

$$\begin{aligned} F(\alpha)' &= 4204\alpha_0 - 557 = 0 \\ \implies \alpha_0 &= 0.13 \end{aligned}$$

$$\begin{aligned} x_1 &= x_0 + \alpha_0 p_0 \\ &= (0, 0) + 0.13(19, 14) = (2.47, 1.82) \end{aligned}$$

$$f_1 = -36.9$$

$$\nabla f_1 = (2.58, -4.25)$$

$$\begin{aligned} \beta_1 &= \frac{\nabla f_1^T \nabla f_1}{\nabla f_0^T \nabla f_0} \\ &= 0.04 \end{aligned}$$

$$\begin{aligned} p_1 &= -\nabla f_1 + \beta_1 p_0 \\ &= (-2.58, 4.25) + 0.04(19, 14) = (-1.82, 4.81). \end{aligned}$$

iteration 2:

$$\begin{aligned} F(\alpha_1) &= 4(2.47 - 1.82\alpha_1)^2 + (2.47 - 1.82\alpha_1)(1.82 + 4.81\alpha_1) + 2(1.82 + 4.81\alpha_1)^2 - 19(2.47 - 1.82\alpha_1) - 14(1.82 + 4.81\alpha_1) \\ &= 50.7676\alpha_1^2 - 25.14\alpha_1 - 36.9 \end{aligned}$$

$$\begin{aligned} F(\alpha_1)' &= 101.54\alpha_1 - 25.14 = 0 \\ \implies \alpha_1 &= 0.25 \end{aligned}$$

$$\begin{aligned} x_2 &= x_1 + \alpha_1 p_1 \\ &= (2.47, 1.82) + 0.25(-1.82, 4.81) = (2.02, 3.02) \end{aligned}$$

$$f_2 = -40$$

$$\nabla f_2 = (0.18, 0.1)$$

$$\begin{aligned} \beta_2 &= \frac{\nabla f_2^T \nabla f_2}{\nabla f_1^T \nabla f_1} \\ &= 0.0017 \end{aligned}$$

$$\begin{aligned} p_2 &= -\nabla f_2 + \beta_2 p_1 \\ &= (-0.18, -0.1) + 0.0017(-1.82, 4.81) = (-0.18, -0.092). \end{aligned}$$

iteration 3:

$$\begin{aligned} F(\alpha_2) &= 4(2.02 - 0.18\alpha_2)^2 + (2.02 - 0.18\alpha_2)(3.02 - 0.092\alpha_2) + 2(3.02 - 0.092\alpha_2)^2 - 19(2.02 - 0.18\alpha_2) - 14(3.02 - 0.092\alpha_2) \\ &= 0.16\alpha_2^2 - 0.04\alpha_2 - 40 \end{aligned}$$

$$\begin{aligned} F(\alpha_2)' &= 0.32\alpha_2 - 0.04 = 0 \\ \implies \alpha_2 &= 0.125 \end{aligned}$$

$$\begin{aligned}x_3 &= x_2 + \alpha_2 p_2 \\ &= (2.02, 3.02) + 0.125(-0.18, -0.092) = (2, 3)\end{aligned}$$

$$f_3 = -40.$$

**Comments:**

Since  $F(x)$  is a quadratic function, then we expect exact convergence in  $n = 2$  iterations (where  $n$  is the dimensionality).

The actual minimum  $F^*(x) = -40$  occurs at  $x^* = (2, 3)$ .

Our solution did indeed converge to  $x^*$  (if not for the round-off errors) in the second iteration. The third iteration removed the round-off errors, and achieved exact convergence.

**ii.**  $F(x) = x_1^2 x_2^2 + (x_1 - 3)^2 + (x_2 - 2)^2$

$$\begin{aligned}F_{x_1} &= 2x_1 x_2^2 + 2(x_1 - 3) \\ F_{x_2} &= 2x_1^2 x_2 + 2(x_2 - 2) \\ \nabla F &= (2x_1 x_2^2 + 2(x_1 - 3), 2x_1^2 x_2 + 2(x_2 - 2)).\end{aligned}$$

iteration 1:

$$\begin{aligned}x_0 &= (0, 0) \\ f_0 &= 13\end{aligned}$$

$$\nabla f_0 = (-6, -4)$$

$$p_0 = -\nabla(f_0) = (6, 4)$$

$$\begin{aligned}F(\alpha_0) &= (6\alpha_0)^2(4\alpha_0)^2 + (6\alpha_0 - 3)^2 + (4\alpha_0 - 2)^2 \\ &= 576\alpha_0^4 + 52\alpha_0^2 - 52\alpha_0 + 13\end{aligned}$$

$$\begin{aligned}F(\alpha)' &= 2304\alpha_0^3 + 104\alpha_0 - 52 = 0 \\ \implies \alpha_0 &= 0.23 \quad (\text{This is the only real solution, the remaining 2 solutions are complex.})\end{aligned}$$

$$\begin{aligned}x_1 &= x_0 + \alpha_0 p_0 \\ &= (0, 0) + 0.23(6, 4) = (1.38, 0.92)\end{aligned}$$

$$f_1 = 5.40$$

$$\nabla f_1 = (-0.90, 1.34)$$

$$\begin{aligned}\beta_1 &= \frac{\nabla f_1^T \nabla f_1}{\nabla f_0^T \nabla f_0} \\ &= 0.05\end{aligned}$$

$$\begin{aligned}p_1 &= -\nabla f_1 + \beta_1 p_0 \\ &= (0.90, -1.34) + 0.05(6, 4) = (1.2, -1.14).\end{aligned}$$

iteration 2:

$$\begin{aligned} F(\alpha_1) &= (1.38 + 1.2\alpha_1)^2(0.92 - 1.14\alpha_1)^2 + (1.38 + 1.2\alpha_1 - 3)^2 + (0.92 - 1.14\alpha_1 - 2)^2 \\ &= 1.87\alpha_1^4 + 1.28\alpha_1^3 - 0.51\alpha_1^2 - 2.62\alpha_1 + 5.40 \end{aligned}$$

$$\begin{aligned} F(\alpha_1)' &= 7.48\alpha_1^3 + 3.84\alpha_1^2 - 1.02\alpha_1 - 2.62 = 0 \\ \implies \alpha_1 &= 0.62 \end{aligned}$$

$$\begin{aligned} x_2 &= x_1 + \alpha_1 p_1 \\ &= (1.38, 0.92) + 0.62(1.2, -1.14) = (2.12, 0.21) \end{aligned}$$

$$f_2 = 4.18$$

$$\nabla f_2 = (-1.57, -1.69)$$

$$\begin{aligned} \beta_2 &= \frac{\nabla f_2^T \nabla f_2}{\nabla f_1^T \nabla f_1} \\ &= 2.04 \end{aligned}$$

$$\begin{aligned} p_2 &= -\nabla f_2 + \beta_2 p_1 \\ &= (1.57, 1.69) + 2.04(1.2, -1.14) = (4.02, -0.64). \end{aligned}$$

iteration 3:

$$\begin{aligned} F(\alpha_2) &= (2.12 + 4.02\alpha_2)^2(0.21 - 0.64\alpha_2)^2 + (2.12 + 4.02\alpha_2 - 3)^2 + (0.21 - 0.64\alpha_2 - 2)^2 \\ &= 6.62\alpha_2^4 + 2.64\alpha_2^3 + 14.54\alpha_2^2 - 5.24\alpha_2 + 4.18 \end{aligned}$$

$$\begin{aligned} F(\alpha_2)' &= 26.48\alpha_2^3 + 7.92\alpha_2^2 + 29.08\alpha_2 - 5.24 = 0 \\ \implies \alpha_2 &= 0.17 \end{aligned}$$

$$\begin{aligned} x_3 &= x_2 + \alpha_2 p_2 \\ &= (2.12, 0.21) + 0.17(4.02, -0.64) = (2.80, 0.10) \end{aligned}$$

$$f_3 = 3.73.$$

**Comments:**

Since  $F(x)$  is a quartic function (and, thus not a quadratic), then we don't expect convergence in  $n$  iterations (where  $n$  is the dimensionality).

The actual minimum  $F^*(x) = 3.58$  occurs at  $x^* = (2.86, 0.22)$ .

The distance from  $x_3$  to  $x^*$  (in the Euclidean 2-norm) is:  $\|x_3 - x^*\|_2 = 0.134 = 13.4\%$ .

This is a large error.

Conclusion: More iterations are required for convergence.

## 2.

(i). Certify that the following Davidon-Fletcher-Powell (DFP) updating-formula satisfies the condition for a Quasi-Newton method.

$$A_{k+1} = A_k - \frac{A_k y_k y_k^T A_k}{y_k^T A_k y_k} + \frac{s_k s_k^T}{s_k^T y_k}$$

where,

$$A_k \approx H^{-1}, \quad s_k = x_{k+1} - x_k, \quad y_k = g_{k+1} - g_k$$

(ii). Hence, carry out two steps to minimize  $F(x)$  of problem (1)(i) using calculus to find the minimum in each search direction.

(iii). Compare with the updating formula given by:

$$B_{k+1} = B_k + \frac{z z^T}{z^T s_k}$$

where,

$$z = y_k - B_k s_k, \quad B_k \approx H.$$

### Solution:

**i. Show that  $A_{k+1}$  satisfies the Quasi-Newton Secant Equation:  $A_{k+1} y_k = s_k$**

*Proof.*

$$A_{k+1} = A_k - \frac{A_k y_k y_k^T A_k}{y_k^T A_k y_k} + \frac{s_k s_k^T}{s_k^T y_k}$$

multiply by  $y_k$  on the right:

$$A_{k+1} y_k = A_k y_k - \frac{A_k y_k y_k^T A_k y_k}{y_k^T A_k y_k} + \frac{s_k s_k^T y_k}{s_k^T y_k}$$

Group:

$$A_{k+1} y_k = A_k y_k - \frac{A_k y_k (y_k^T A_k y_k)}{(y_k^T A_k y_k)} + \frac{s_k (s_k^T y_k)}{(s_k^T y_k)}$$

Simplify:

$$A_{k+1} y_k = A_k y_k - A_k y_k + s_k = s_k.$$

□

ii. **Minimize:**  $F(x) = 4x_1^2 + x_1x_2 + 2x_2^2 - 19x_1 - 14x_2$

We'll use the BFGS algorithm, which is given by:

**Algorithm 6.1** (BFGS Method).

Given starting point  $x_0$ , convergence tolerance  $\epsilon > 0$ ,  
inverse Hessian approximation  $H_0$ ;

$k \leftarrow 0$ ;

**while**  $\|\nabla f_k\| > \epsilon$ ;

    Compute search direction

$$p_k = -H_k \nabla f_k;$$

    Set  $x_{k+1} = x_k + \alpha_k p_k$  where  $\alpha_k$  is computed from a line search  
    procedure to satisfy the Wolfe conditions (3.6);

    Define  $s_k = x_{k+1} - x_k$  and  $y_k = \nabla f_{k+1} - \nabla f_k$ ;

    Compute  $H_{k+1}$  by means of (6.17);

$k \leftarrow k + 1$ ;

**end (while)**

$$\begin{aligned} F_{x_1} &= 8x_1 + x_2 - 19 \\ F_{x_2} &= x_1 + 4x_2 - 14 \\ \nabla F &= (8x_1 + x_2 - 19, x_1 + 4x_2 - 14). \end{aligned}$$

iteration 1:

$$x_0 = (0, 0)$$

$$f_0 = 0$$

$$A_0 = I$$

$$\nabla f_0 = (-19, -14)$$

$$\begin{aligned} p_0 &= -A_0 \nabla f_0 \\ &= -I(-19, -14) = (19, 14) \end{aligned}$$

$$\begin{aligned} F(\alpha_0) &= 4(19\alpha_0)^2 + (19\alpha_0)(14\alpha_0) + 2(14\alpha_0)^2 - 19(19\alpha_0) - 14(14\alpha_0) \\ &= 1444\alpha_0^2 + 266\alpha_0^2 + 392\alpha_0^2 - 361\alpha_0 - 196\alpha_0 \\ &= 2102\alpha_0^2 - 557\alpha_0 \end{aligned}$$

$$\begin{aligned} F(\alpha)' &= 4204\alpha_0 - 557 = 0 \\ \implies \alpha_0 &= 0.13 \end{aligned}$$

$$\begin{aligned}x_1 &= x_0 + \alpha_0 p_0 \\&= (0, 0) + 0.13(19, 14) = (2.47, 1.82)\end{aligned}$$

$$f_1 = -36.9$$

$$\nabla f_1 = (2.58, -4.25)$$

$$s_0 = x_1 - x_0 = (2.47, 1.82)$$

$$y_0 = g_1 - g_0 = (21.58, 9.75)$$

$$\begin{aligned}A_1 &= A_0 - \frac{A_0 y_0 y_0^T A_0}{y_0^T A_0 y_0} + \frac{s_0 s_0^T}{s_0^T y_0} \\&= I - \frac{I \begin{bmatrix} 21.58 \\ 9.75 \end{bmatrix} \begin{bmatrix} 21.58 & 9.75 \end{bmatrix} I}{\begin{bmatrix} 21.58 & 9.75 \end{bmatrix} I \begin{bmatrix} 21.58 \\ 9.75 \end{bmatrix}} + \frac{\begin{bmatrix} 2.47 \\ 1.82 \end{bmatrix} \begin{bmatrix} 2.47 & 1.82 \end{bmatrix}}{\begin{bmatrix} 2.47 & 1.82 \end{bmatrix} \begin{bmatrix} 21.58 \\ 9.75 \end{bmatrix}} \\&= I - \frac{\begin{bmatrix} 465.70 & 210.41 \\ 210.41 & 95.10 \end{bmatrix}}{560.76} + \frac{\begin{bmatrix} 6.10 & 4.50 \\ 4.50 & 3.31 \end{bmatrix}}{71.1} \\&= I - \begin{bmatrix} 0.83 & 0.38 \\ 0.38 & 0.17 \end{bmatrix} + \begin{bmatrix} 0.09 & 0.06 \\ 0.06 & 0.05 \end{bmatrix} \\&= \begin{bmatrix} 0.26 & -0.32 \\ -0.32 & 0.88 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}p_1 &= -A_1 \nabla f_1 \\&= \begin{bmatrix} -0.26 & 0.32 \\ 0.32 & -0.88 \end{bmatrix} \begin{bmatrix} 2.58 \\ -4.25 \end{bmatrix} \\&= \begin{bmatrix} -2.03 \\ 4.57 \end{bmatrix}\end{aligned}$$

iteration 2:

$$\begin{aligned}F(\alpha_1) &= 4(2.47 - 2.03\alpha_1)^2 + (2.47 - 2.03\alpha_1)(1.82 + 4.57\alpha_1) + 2(1.82 + 4.57\alpha_1)^2 - 19(2.47 - 2.03\alpha_1) - 14(1.82 + 4.57\alpha_1) \\&= 48.98\alpha_1^2 - 24.66\alpha_1 - 36.89\end{aligned}$$

$$\begin{aligned}F(\alpha_1)' &= 97.96\alpha_1 - 24.66 = 0 \\&\implies \alpha_1 = 0.25\end{aligned}$$

$$\begin{aligned}x_1 &= x_1 + \alpha_1 p_1 \\&= (2.47, 1.82) + 0.25(-2.03, 4.57) = (1.96, 2.96)\end{aligned}$$

$$f_1 = -39.99.$$

**iii. Compare the Update Formulas  $A_{k+1}$  and  $B_{k+1}$ :**